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DESIGN, IMPLEMENTATION AND UTILIZATION OF DIGITAL REPOSITORY SYSTEM FOR STUDENTS OF DEPARTMENT OF SCIENCE EDUCATION, ATBU BAUCHI

Hassan Daiyabu[1], Babawuro Shuaibu[2], & Musa Olarewaju Dauda[3]

[1]Department of Science Education ATBU Bauchi
[2]Department of Vocational and Technical Education ATBU Bauchi
[3]Department of Science Education ATBU Bauchi

Faculty of Technology Education, Abubakar Tafawa Balewa University,
P.M.B. 0248 Bauchi. Nigeria

Corresponding Author: aldaiyab1@gmail.com +2348060095946

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Abstract

The study is design, implementation and utilization of a digital repository system for students in the Department of Science Education at ATBU Bauchi. During design and implementation, application development tools such as HTML, PHP, and CSS were used, and the digital repository system was hosted online at (https://www.atbudserepository.com.ng). The study is guided by four specific objectives and four research questions. A survey design was adopted for the study. The population of the study comprised 1,678 students of Department of Science Education, Faculty of Technology Education, ATBU Bauchi, from whom 313 sample respondents were drawn using Convenience sampling techniques. Instruments for data collection used were three electronic questionnaires, titled Design Features Questionnaire for Development of Digital Repository (DFQDDR), Digital Repository System Development Procedure Questionnaire (DRSDPQ), and Digital Repository Usability Questionnaire (DRUQ). The questionnaires were validated by two Measurement and Evaluation experts and two computer scientist and all observations made were effected. The data collected were coded and analyzed using descriptive statistics (mean and standard deviation). The findings revealed that the digital repository system can be design and implemented using Software development tools such as PHP, HTML and JavaScript. Student's usability was very satisfactory with a mean score above 3.00 and Software Development Models and tools used were appropriate and developed digital repository system is suitable for deployment and usage and it can serve the intended purpose. Recommendations were made such as approval of the Digital Repository System in the Department for usage and Liaison with Directorate of Information Communication Technology of the University to append the Digital Repository link on the University Portal for more usage.

Keywords: ATBU Bauchi, Science Education, Utilization, Digital Repository

Introduction

Digital repositories are a popular technology that universities and other research institutions use to increase their global visibility, intellectual resource documentation, intellectual resource availability, accessibility, and usability for the research community. A university performance is determined by the quality of research conducted and the availability, accessibility, and usability of that research by interested bodies and individuals outside the university. Most importantly, a digital repository would serve as an intersection of teaching, learning, research, and digital library environments (Koto, Syukri & Arief, 2018).

Abubakar Tafawa Balewa University, Bauchi (ATBU) is a Federal University of Technology with 7 faculties offering courses in 42 department(s) and degrees offered in about 399 areas. The digital transformation of traditional products towards highly interdisciplinary and connective smart products, including related internet-based services, brings changes in the availability, accessibility, and usability of intellectual resources (Eickhoff, Aiden, Gobel & Eigner, 2020). Similarly, Asadi, Abdullah, Yah and Sidi, (2019) said Digital repositories have potentially increased the public value, ranking, prestige, and visibility of researchers and universities.

A digital repository is a mechanism for managing and storing digital content based on availability, accessibility, and usability. Putting content into an institutional repository enables staff, students, and institutions to manage and preserve it. The goal of university digital repositories worldwide is to provide open access to their collections (Abdelrahman, 2021).

The design process was the waterfall model. The waterfall model is a traditional system development life cycle (SDLC) that is widely used across many fields of project development. SDLC is an approached designed by professionals to ensure project get positive result at the end. Waterfall on the other hand is simpler to implement and less expensive, that being the reason it is more widely used (Adetokunbo & Basirat, 2014).

Nwachi and Idoko (2021) are of the opinion that digital repositories can advance a surprising number of goals and address an impressive range of needs. Consistent utilization of the digital repository by students, academics, and the university community is a key factor that would help and contribute to achieving the goals and objectives. The researcher intends to investigate how Department of science education students are utilizing the designed digital repository in their learning and research activities. Bharathi and Sujatha (2019) stated that it's important to conduct the study to know the use of digital information resources by the faculty and students.

Statement of the Problem

Universities and other research institutions globally are now competing for the use of sophisticated infrastructure that can ease their mode of operation, be it academics or administration, in order to enhance their visibility among others. It's evident that automating some form of university process, such as archiving intellectual resources and other academic documents, may definitely enhance student productivity in the areas of teaching and learning, research and development (R&D).

Department of Science Education, ATBU Bauchi, used a traditional method of archiving students' intellectual resources (their research works), which was not automated. The method has a lot of disadvantages. Okon, Eleberi and Uka (2020) outlined numerous problems associated with the traditional system of storing scholarly materials in tertiary institutions. Among are: users who need access to these materials must physically go to the library or offices, which can often lead to congestion; the inability for intellectual resources to be accessed at any time (24/7); the fact that intellectual resources cannot be used concurrently; limited storage space; the degradation of hard copy scholarly materials from repeated use and high cost; and stress and time wasted in publishing on external journal repositories. Another problem associated with the traditional system is that it hinders the availability, accessibility, and usability of the intellectual resources (student research works), which impacts both internal and external research users negatively. In the digital age, the amount of data produced is growing exponentially. Governments and institutions can no longer rely on old methods for storing data and passing on knowledge to future generations. Digital data preservation is a mandatory issue that needs proper strategies and tools (Rosa, Craveiro, & Domingues, 2017).

The study aims to design, implement, and use a digital repository system for students of the Science Education Department at the University of Bauchi in order to address the aforementioned issues. The implementation of a web-based digital repository for scholarly materials and publications will help curb the problems listed above by providing an efficient and reliable solution for prompt uploading, storing, and retrieval of the institution's scholarly materials. Also, being web-based means that you can have several users access the repository and archived materials simultaneously.

Purpose of the Study

The main purpose of the study is to Design, Implement and determine students' utilization of the Digital Repository System for Department of Science Education Students of ATBU Bauchi. Waterfall model of software development was a guide to the design and the specific objectives are:

i. To determine the suitable features for the design of Digital Repository System for Students' of Department of Science Education, ATBU Bauchi.
ii. To identify the suitable procedures for development of Digital Repository System for Students' of Department of Science Education, ATBU Bauchi.
iii. To determine the Users' Satisfaction of the Digital Repository System for Students' of Department of Science Education, ATBU Bauchi

Research Questions

Three (3) research questions guided the study:

i. What are the features appropriate for the design of Digital Repository System for Students' of Department of Science Education, ATBU Bauchi?
ii. What are the suitable procedures for the development of Digital Repository System for Students' of Department of Science Education, ATBU Bauchi?
iii. What is the Users' Satisfaction of the Digital Repository System for Students' of Department of Science Education, ATBU Bauchi?

Literature Review

Digital repository is a collection of digital objects stored in an sustainable, reliable and trusted environment which manages both content and metadata, enables the deposition of content by multiple users, offers a set of basic services. Minkova (2018) further outlined that Despite these broad characteristics are useful for framing the term, further qualification is required with regards to the contents and the purpose of the repository.

Koumoutsos, Mitrelis and Tsakonas (2010) described and considered Digital repositories are essential information tools for scholarly communication, where it acceptability and extensive use by communities and institutions, as well as the users' commitment in self- archiving, highlight the need for developing alternative channels of communication to expose scholarly productivity. Furthermore, the digital repositories developers are boldly interested into transforming the trends into viable, reliable and useful systems with any idea of computer programming knowledge.

Digital repositories are becoming more and more popular because it does facilitate the processes of depositing and storing of digital content. Via them educational institutions and lecturers can share their intellectual production in an environment that provides powerful tools for its management. Kiryakova and Yordanova (2013). A digital repository is a mechanism for managing and storing digital content on the major three aspect accessibility, availability and usability.

Recently, educational organisations and university ranking bodies includes Institutional repository as one of the major performance indicator where Universities and other educational institutions is been ranked. This assertion was supported by Tsunoda, Sun, Nishizawa and Liu (2016) in their article while accessing the Academic and Research Impact of Shared Contents in Institutional Repositories in Related to Performance Indicators of University Rankings.

Similarly, Kubik and Kwiecień (2021) opined that, construction of a digital repository is a task combining elements related to the IT project management (ending with a software system implementation) and activities connected with the acquisition and publication of digital resources (daily use of the implemented system).

Bassil (2015) outlined that there exist many SDLC models, one of which is the waterfall model, which comprises five, phases to be completed sequentially in order to develop a software solution. Khan (2020) discoursed that the main aim of SDLC is to produce high-quality software that meets customer expectations. . Khan (2020) also refers to SDLC as Application Development Life Cycle, which is not a methodology but a description of various phases that are involved in software development, starting from project definition to deployment and sustainment. These SDLC phases serve as a programmatic guide to project activity. The SDLC models are (Waterfall, Spiral, V-Model, Iterative, Big Bang, Agile, and Rapid Application Model). Furthermore, the study outlined some advantages and disadvantages of the waterfall model.

Nwokedi and Nwokedi (2018) affirmed that establishment of Institutional Repositories in academic and research institutions in Nigeria are a serious developmental issue that requires urgent attention.

Methodology

Descriptive survey design was adopted for the study. The population of the study consisted of 1,678 Department of Science Education Students both postgraduate and undergraduate. A convenience sampling technique was used. The sample size was 313 Students drawn across all courses. Questionnaire was used for data collection titled Design Features Questionnaire for Development of Digital Repository (DFQDDR), Digital Repository System Development Procedure Questionnaire (DRSDPQ) and Digital Repository Usability Questionnaire (DRUQ) all the questionnaires were faced validated by experts. Cronbach alpha reliability technique was used to determine the reliability of the instrument which yielded a reliability coefficient of .703, .701 and 0.80 respectively. The instrument was designed using a 5 point Likert scale Very Highly Suitable (VHS), Highly Suitable (HS), Moderately Suitable (MS), Lowly Suitable (LS) and Not Suitable (NS). Mean and standard deviation was used as a method of data analysis.

Result and Discussion

The tables below are the results of analysed data presented for the purpose of answering all the three research questions in the study.

Table 1: Mean responses of computer programmers on suitable programming language adopted for design of Digital Repository System.

S/N	Items	Mean	SD	Remark
1.	Hypertext markup language version 5 (HTML5)	4.31	0.75	Highly Suitable
2.	Cascading style sheet version 3 (CSS 3)	4.00	1.22	Highly Suitable
3.	Java	3.92	1.18	Highly Suitable
4.	Node-Js	3.00	1.52	Moderately Suitable
5.	Hypertext preprocessor (PHP)	4.46	0.77	Highly Suitable
6.	JavaScript	3.46	1.50	Moderately Suitable
7.	C++	3.85	1.40	Highly Suitable
	Grand Mean	**3.85**	**1.19**	**Highly Suitable**

Source: Field survey 2022

Key: Very Highly Suitable = VHS, Highly Suitable = HS, Moderately Suitable = MS, Lowly Suitable = LS, Not Suitable = NS.

Table 1 on research question 1 revealed that 7 items with grand mean 3.85 and individual items had their mean values ranging from 3.00 to 4.46. Whereas items number 1, 2 and 5 with mean of 4.31, 4.00 and 4.46 respectively. This indicate that, the three programming language were the most highly suitable for design of the DRS. The Table also showed that the standard deviations (SD) of the items are within the range of 0.75, 1.22 and 0.77 all are positive. This indicated that the respondents were not very far from the Mean or from one another in their responses.

Table 2: Mean responses of computer programmers on suitable Database and development environment adopted for design of Digital Repository System.

S/N	Items	Mean	SD	Remark
1.	MySQL	4.62	0.65	Very Highly Suitable
2.	Oracle	3.15	0.89	Moderately Suitable
3.	PostgreSQL	3.38	1.26	Moderately Suitable
4.	Microsoft Azure SQL Database	3.15	1.28	Moderately Suitable
5.	DbVisualizer	3.31	1.03	Moderately Suitable
6.	MangoDb	4.08	1.11	Highly Suitable
	Grand Mean	**3.62**	**1.03**	**Highly Suitable**

Source: Field survey 2022

Key: Very Highly Suitable = VHS, Highly Suitable = HS, Moderately Suitable = MS, Lowly Suitable = LS, Not Suitable = NS.

Table 2 on research question 1 revealed that 6 items with grand mean 3.62 and individual items had their mean values ranging from 3.15 to 4.62. This indicated that all the 6 databases development environments were suitable for the design of DRS. Standard deviations (SD) of the items are within the range of 0.65 to 1.261 and are positive. Whereas items number 1 and 6 with mean of 4.62 and 4.08 respectively, the two items has the highest suitability and prepared databases for the design of the DRS. This indicated that the respondents were not very far from the Mean or from one another in their responses.

Table 3: Mean responses of computer programmers on suitable components and interface design adapted for design of the Digital Repository System.

S/N	Items	Mean	SD	Remark
1.	Work on desktop and mobile devices.	4.23	0.83	Highly Suitable
2.	Accessibility only by authorized/registered User-name and password.	4.00	1.00	Highly Suitable
3.	Authorized/Registered users ease of access to the web-server.	4.08	0.95	Highly Suitable
4.	Restricted user performance.	3.92	1.25	Highly Suitable
5.	Students ease of DRS use keyboard skills.	3.23	1.23	Moderately Suitable
	Grand Mean	**3.89**	**1.05**	**Highly Suitable**

Source: Field survey 2022

Key: Very Highly Suitable = VHS, Highly Suitable = HS, Moderately Suitable = MS, Lowly Suitable = LS, Not Suitable = NS.

Table 3 on research question 1 also revealed that 5 items on components and interface design adopted for design of the Digital Repository System had the grand mean 3.89 and individual items also has their mean values ranging from 3.23 to 4.23. Standard deviations (SD) of the items 2, 4 and 5 are within the range of 1.00 to 1.25 and items number 1 and 3 standard deviations (SD) were 0.83 and 0.95 respectively. This indicated that the respondents agreed all the items were highly suitable components for the design of interface of the digital repository system.

Table 4: Mean responses of computer programmers on suitable development procedure adopted for development of Digital Repository System.

S/N	Items	Mean	SD	Remark
1.	Brainstorming and planning	4.38	0.76	Highly Suitable
2.	Requirements and feasibility analysis	4.31	0.85	Highly Suitable
3.	Design	4.08	0.95	Highly Suitable
4.	Development and coding	4.15	0.55	Highly Suitable
5.	Development Ops Methodology	3.31	0.75	Moderately Suitable
6.	Agile Software Development Methodology	3.31	1.18	Moderately Suitable
7.	Integration and testing	3.92	1.03	Highly Suitable
8.	Waterfall Model	4.00	1.00	Highly Suitable
9.	Feature Driven Development	3.54	1.33	Highly Suitable
10.	Prototype Methodology	3.85	1.21	Highly Suitable
11.	Rapid Application Development (RAD)	3.85	0.68	Highly Suitable
12.	Spiral Model	3.46	1.39	Moderately Suitable
13.	Joint Application Development Methodology	3.54	1.19	Highly Suitable
	Grand Mean	**3.80**	**0.99**	**Highly Suitable**

Source: Field survey 2022

Key: Very Highly Suitable = VHS, Highly Suitable = HS, Moderately Suitable = MS, Lowly Suitable = LS, Not Suitable = NS.

Table 4 on research question 2 revealed that 13 items are with the grand mean 3.80 and individual items had their mean values ranging from 3.31 to 4.38. This indicated that, all 10 methodologist and their procedures are within 3.54 to 4.38 (highly suitable) the remaining 3 are from 3.31 to 3.54 (Suitable) for the development of DRS. Standard deviations

(SD) of the items are within the range of 0.55 to 1.39 and are positive. This indicated that the respondents agreed all the above development procedure adopted for development of Digital Repository System were highly suitable.

Table 5: Mean responses of Users' Satisfaction of the Digital Repository System suitability.

S/N	Items	Mean	SD	Remark
1.	The repository is easy to navigate	3.22	1.17	Moderately Suitable
2.	The search functionality is effective in finding relevant materials	3.12	1.10	Moderately Suitable
3.	The interface design is user-friendly	3.35	1.19	Moderately Suitable
4.	The materials in the repository is relevant and appropriate for the target audience	3.20	1.28	Moderately Suitable
5.	The content was accurate and up-to-date	3.24	1.12	Moderately Suitable
6.	The content presented in a way that is easy to understand and use	3.19	1.14	Moderately Suitable
7.	The repository is meeting the needs and expectations of its users?	3.38	1.17	Moderately Suitable
8.	The repository receives feedback from users and there are some complaints or issues.	3.41	1.13	Moderately Suitable
	Grand Mean	**3.26**	**1.16**	**Moderately Suitable**

Source: Field survey 2022
Key: Very Highly Suitable = VHS, Highly Suitable = HS, Moderately Suitable = MS, Lowly Suitable = LS and Not Suitable = NS

Table 5 research question 3 revealed that 8 items on the Users' Satisfaction of the Digital Repository System are with the grand mean 3.26 and individual items had their mean values ranging from 3.12 to 3.41, and this indicated that the items were suitable and Users satisfied with the system usability. Standard deviations (SD) of the items also are within the range of 1.10 to 1.28 and are positive. This indicated that the respondents agreed and satisfied that digital repository system is capable and met all the expected requirements.

Discussion of Findings

Finding of the study on suitable programming language adopted for design of Digital Repository System, indicated that Computer Programmers identified the following Programming and scripting languages to be suitable for designing the Digital Repository System: Hypertext markup language version 5 (HTML 5); Cascading style sheet version 3 (CSS 3); Java; Node-Js; Hypertext preprocessor (PHP); JavaScript and C++. This is justified by Tosin (2015) which used in designing and implementing a computerized library management system. The library management system was designed and implemented using HTML (Hypertext Markup Language), CSS (Cascading Style Sheets), PHP (Hypertext Pre-Processor) and Iwayemi and Oyeniyi (2019) in their study, "Development of a Robust Library Management System," The design was aimed at providing solutions to various challenges faced by the manual library system.

The result of data analysis in Tables 2 for research question 1 on Suitable Database and development environment adopted for design of Digital Repository System. It was indicated that computer programmers identified the following: MySQL; Oracle; PostgreSQL; Microsoft Azure SQL Database; DbVisualizer and MangoDb were suitable to be used in developing the Digital Repository System backend. This findings is in agreement with the study conducted by Tosin (2015) and Iwayemi and Oyeniyi (2019) which acknowledged that such system is developed using database software like MySQL at the backend. Also Lian and Wang (2021) reported that relational databases have gained widespread attention in the whole industry.

The result of data analysis in table 4 research question 2 on suitable procedures for development of Digital Repository System for Students' of Department of Science Education, ATBU Bauchi. It was outlined by the computer programmers agreed that, Brainstorming and planning; Requirements and feasibility analysis; Design; Development and coding; Agile Software Development Methodology; Integration and testing; Waterfall Model; Prototype Methodology; Rapid Application Development (RAD); Spiral Model; Joint Application Development Methodology and their procedures were suitable for adoption in the design of the Digital Repository System. This findings is in agreement with Khan (2020); Seimiekumo Solomon *et al.* (2019); Wang and Zhang (2013), suggested these development methodologies and their procedures were appropriate for development of Software and online system.

The analysis of table 5 for the research question 3 on the Users' Satisfaction of the Digital Repository System for Students' of Department of Science Education, ATBU Bauchi. This indicated that Students of the Department of Science Education agreed that the digital repository system is easy to navigate; search functionality is effective; interface design is user – friendly; materials in the repository is relevant and appropriate to the target audience; also they agreed to some extent that contents are up to date; contents is presented in an understandable manner and it also meet the expectation of users. These findings is in conformity with the findings of Akparobore and Omosekejimi (2020) which stated that users attitude towards Institution Repository were very positive and accessing contents for their academic improvement. Bharathi and Sujatha (2019) revealed that user's dependency on digital information resources is positively increased in their activities and has an impact on their academic pursuit. Wangui (2018) acknowledged that contents of the institutional repository benefiting students.

Conclusion and Recommendations

The results of this study established a high degree of validity for the package in all areas of investigation through each step of the Digital Repository System development life cycle. The findings also showed that Software Development Models and Application Development Tools such as PHP, HTML, CSS and JavaScript can be used for designing and implementing Digital Repository System. Students Utilization of the Digital Repository System was impactful where it eases and facilitates learning and carrying out Research and Assignment. The following recommendations were made:

i. The Department of Science Education should approve and continue to use the digital repository system and make it more functional to sustain and achieve the desired objectives.
ii. The Department of Science Education should liaise with the University's Directorate of Information and Communication Technology (DICT) to append the digital repository URL (https://www.atbudserepository.com.ng) on the university website for more patronage. Also change the domain name to .edu.ng to make it more official.
iii. The Department of Science Education should appoint an academic staff to be as an administrator that handles uploads and plagiarism checks and deal with any issues that arise. Also, make it mandatory for all the students of the Department to create a user account with the Students' Digital Repository System in order to have more users.

References

Abdelrahman, O. H. (2021). A Study of the Degree of Openness of University Digital Repositories in Sudan. *Asian Journal of Sociological Research*, *4*(1), 1–6.

Adetokunbo, A., & Basirat, A. (2014). Software Engineering Methodologies: A Review of the Waterfall Model and Object-Oriented Approach. *International Journal of Scientific & Engineering Research*, *4*(7), 427–434.

Akparobore, D. O., & Omosekejimi, A. F. (2020). Faculty Members Awareness and Attitude towards the Use of Institutional Repositories (IRs) in Federal Universities in South South, Nigeria. *Library Progress (International)*, *40*(2), 154–165. https://doi.org/10.5958/2320-317x.2020.00018.5

Asadi, S., Abdullah, R., Yah, Y., & Nazir, S. (2019). Understanding Institutional Repository in Higher Learning Institutions: A Systematic Literature Review and Directions for Future Research. *IEEE Access*, *7*(March), 35242–35263. https://doi.org/10.1109/ACCESS.2019.2897729

Bassil, Y. (2015). A Simulation Model for the Spiral Software Development Life Cycle. *International Journal of Innovative Research in Computer and Communication Engineering*, *03*(05), 3823–3830. https://doi.org/10.15680/ijircce.2015.0305013

Bharathi, K., & Sujatha, H. R. (2019). Utilization of the Digital Information Resources and Services Provided in the Engineering College Libraries in Karnataka, India: User's Perspective. *Library Philosophy and Practice (e-Journal)*, *2373*.

Eickhoff, T., Eiden, A., Göbel, J. C., & Eigner, M. (2020). A Metadata Repository for Semantic Product Lifecycle Management. *Procedia CIRP Design Conference 2020*, *91*, 249–254. https://doi.org/10.1016/j.procir.2019.11.006

Iwayemi, A., & Oyeniyi, S. (2019). Development of a Robust Library Management System. *International Journal of Computer Applications*, *178*(12), 9–16. https://doi.org/10.5120/ijca2019918850

Khan, M. E. (2020). Empirical Study of Software Development Life Cycle and its Various Models. *International Journal of Software Engineering (IJSE)*, *8*(2), 16–26.

Kiryakova, G., & Yordanova, L. (2013). Digital Repositories – A Tool for Effective Management of Institutional Assets. *Trakia Journal of Sciences*, *11*(4), 377–383.

Koto, J., Syukri, M., & Arief, D. S. (2018). Performance of Universities based on Research Output. *Journal of Ocean, Mechanical and Aerospace*, *61*(1), 6–15.

Koumoutsos, K., Mitrelis, A., & Tsakonas, G. (2010). Evaluation insights to key processes of digital repositories. *Libraries In the Digital Age 2010, Zadar, Croatia, 24-28 June*, 1–11.

Kubik, T., & Kwiecień, A. (2021). Resolving dilemmas arising during design and implementation of digital repository of heterogenic scientific resources. *Applied Sciences (Switzerland)*, *11*(1), 1–19. https://doi.org/10.3390/app11010215

Lian, J. I. E., & Wang, S. (2021). TDRB : An Efficient Tamper-Proof Detection Middleware for Relational Database Based on Blockchain Technology. *IEEE Access*. https://doi.org/10.1109/ACCESS.2021.3076235

Minkova, M. (2018). *Github as a digital library*. University of Boras.

Nwachi, C., & Idoko, N. (2021). Staff Use of Institutional Repositories for Academic Purposes in Nigerian University Libraries: the Librarians' Perspective. *Library Philosophy and Practice*, *2021*(April), 1–22.

Nwokedi, V. C., & Nwokedi, G. I. (2018). Open Access Institutional Repositories in Academic and Research Institutions in Nigeria: A Review of Benefits and Challenges. *International Journal of Academic Library and Information Science*, *6*(8), 242–252. https://doi.org/10.14662/IJALIS2018.036

Okon, R., Eleberi, E. L., & Uka, K. K. (2020). A Web Based Digital Repository for Scholarly Publication. *Journal of Software Engineering and Applications*, *13*(04), 67–75. https://doi.org/10.4236/jsea.2020.134005

Rosa, C. A., Craveiro, O., & Domingues, P. (2017). Open source software for digital preservation repositories: A survey. *ArXiv*, *8*(3). https://doi.org/10.5121/ijcses.2017.8302

Seimiekumo Solomon, F., Kennedy, Z. O., Deinbofa, G., & Godwin, O. O. (2019). Design And Implementation Of Digital Library Management System. A Case Study Of The Niger Delta University, Bayelsa State. *International Journal of Scientific and Research Publications (IJSRP)*, *9*(12), p96112. https://doi.org/10.29322/ijsrp.9.12.2019.p96112

Tosin, A. T. (2015). *Library Management System By Being a Project Report Submitted To the Department of Computer Science , Faculty of Science , Federal University Oye Ekiti , Ekiti State , Nigeria in Partial Fulfilment of the Requirement for the Award of Bachelor of Science (*. 70.

Tsunoda, H., Sun, Y., Nishizawa, M., & Liu, X. (2016). A study on the academic and research impact of shared contents in institutional repositories in related to performance indicators of university rankings. *Proceedings of the Association for Information Science and Technology*, *53*(1), 1–6. https://doi.org/10.1002/pra2.2016.14505301084

Wang, X. Y., & Zhang, Q. S. (2013). Design and implementation of library management system based on .Net. *Advanced Materials Research*, *760–762*(13), 934–937. https://doi.org/10.4028/www.scientific.net/AMR.760-762.934